

UNIT 12

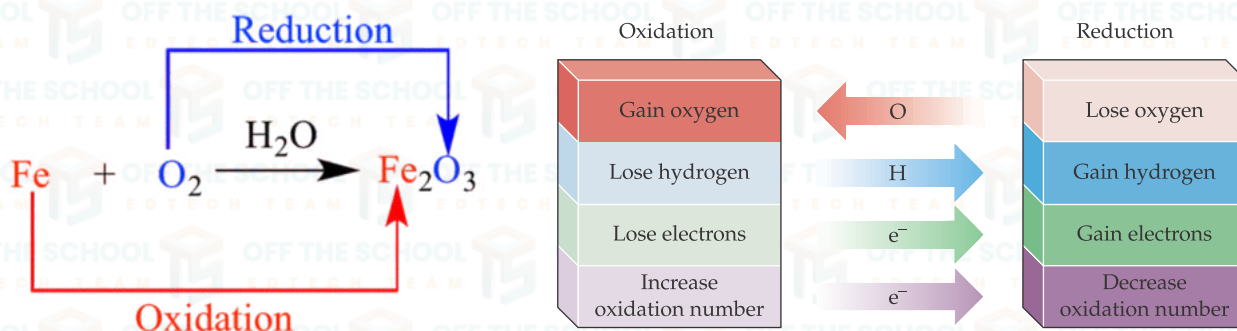
ELECTROCHEMISTRY



OXIDATION	REDUCTION
DEFINITION	
Losing electrons.	Gaining electrons.
OXYGEN INVOLVEMENT	
Originally meant combination with oxygen, but now refers to any loss of electrons.	Initially meant removal of oxygen, but now refers to any gain of electrons.
EXAMPLE	
Iron rusting: $2\text{Fe} + \text{O}_2 \longrightarrow 2\text{FeO}$. Iron (Fe) loses electrons to oxygen, showing oxidation.	Hydrogen reacting with oxygen to form water: $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$. Oxygen is reduced because it gains electrons from hydrogen.
RESULT	
The oxidized substance increases its oxidation state (more positive).	The reduced substance decreases its oxidation state (less positive, more negative).

REDOX REACTION:

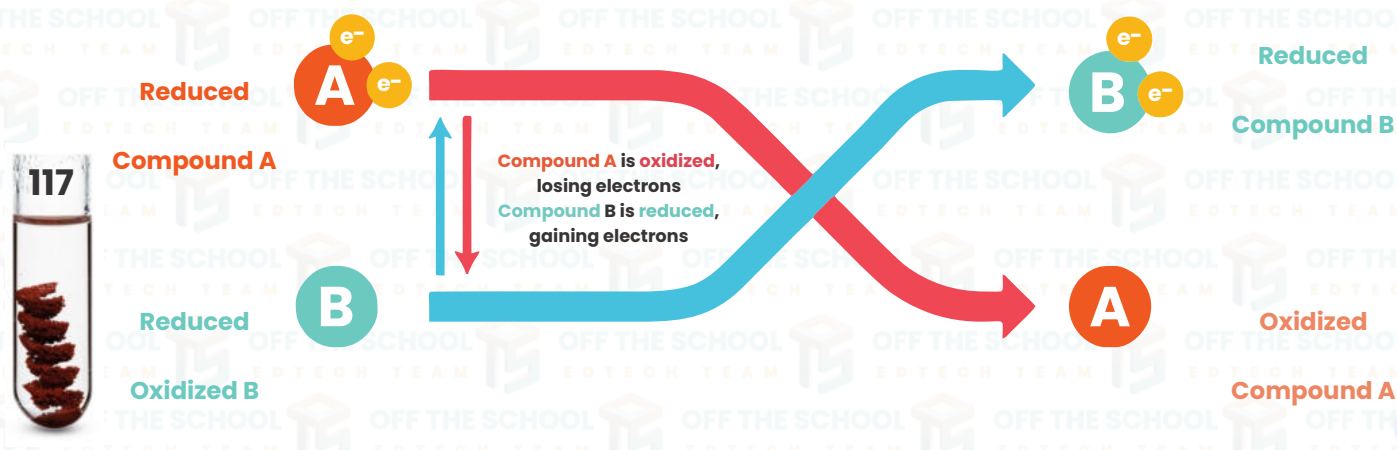
Reduction and oxidation reactions are collectively called as redox reaction.

**OXIDIZING AGENT:**

An oxidizing agent oxidizes another substance – and is itself reduced or simply the chemical species which accepting the electrons. e.g. H_2SO_4 , HNO_3 , KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, Cl_2 , Br_2 , I_2 etc.

REDUCING AGENT:

A reducing agent reduces another substance – and is itself oxidized or simply the chemical species which losing the electrons. e.g., Alkali metals, Al, H_2S , Zn, NaH, KH etc.

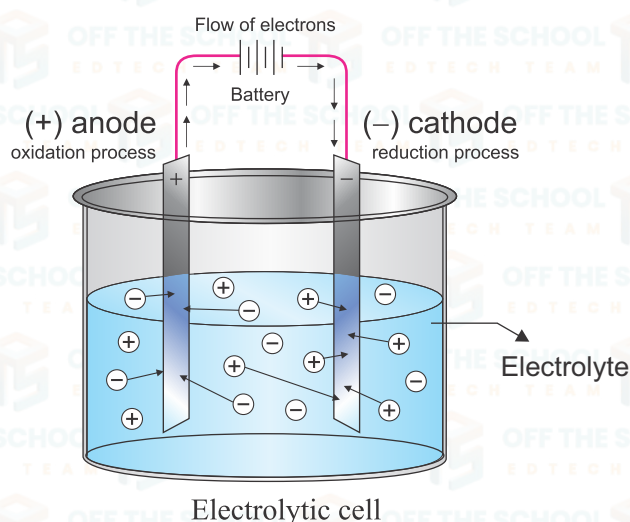


ELECTROLYTIC CELL:

The electrolytic cell is an apparatus containing two electrodes dipped in electrolytic solution & connected to the battery & the current is used to derive a non-spontaneous redox reaction.

Structure:

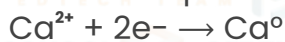
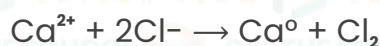
- The electrolytic cell consists of positive & negative electrodes (Electrode is the metallic plate through which electric current enters or leaves the electrolyte).
- They are connected to the battery (source of electricity).
- The electrode, which is connected to the negative terminal is called the cathode.
- The cathode is the electrode through which electrons enter the solution the cathode is the electrode where reduction occurs.
- The electrode, which is connected to the positive terminal, is called the anode.
- The anode is the electrode through which electrons leave the solution. The anode is the electrode where oxidation occurs.
- These electrodes are dipped in an electrolytic solution (an electrolyte is a solution composed of ions, that allow the passage of electric current through them).
- The positive ions are known as cations, while the negative ions are called anions.

**Working:**

- When an electric current is passed through the solution, anions move toward the anode where they lose electrons & get oxidized.
- Cations move towards the cathode where they gain electrons & get reduced.
- The phenomenon is known as Electrolysis.

ELECTROLYSIS:

The movement of cations & anions towards the respective electrodes accompanying a chemical change in an electrolytic solution under influence of electric current.

CHEMICAL REACTION (Electrolysis of CaCl_2):**• Ionization:****• Reduction At cathode:** calcium metal deposited at cathode:**• Oxidation at Anode:** chlorine gas is liberated at anode:**Overall Redox Reaction:**

GALVANIC or VOLTAIC CELL:

The Galvanic or Voltaic cell is a type of electrochemical cell in which current is generated by a spontaneous redox reaction.

Structure:

- The galvanic cell consists of two half-cells which are interconnected electrically by a salt bridge.
- Each half-cell is composed of a metal strip immersed in a solution of its metal ions.
- In the one-half cell, the Zinc electrode is dipped in a solution of Zn^{2+} ions while the other half cell contains a Copper electrode dipped in a solution of Cu^{2+} ions.
- A salt bridge is used to electrically connect two half-cells.

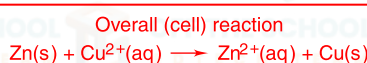
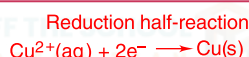
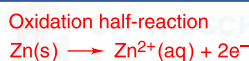
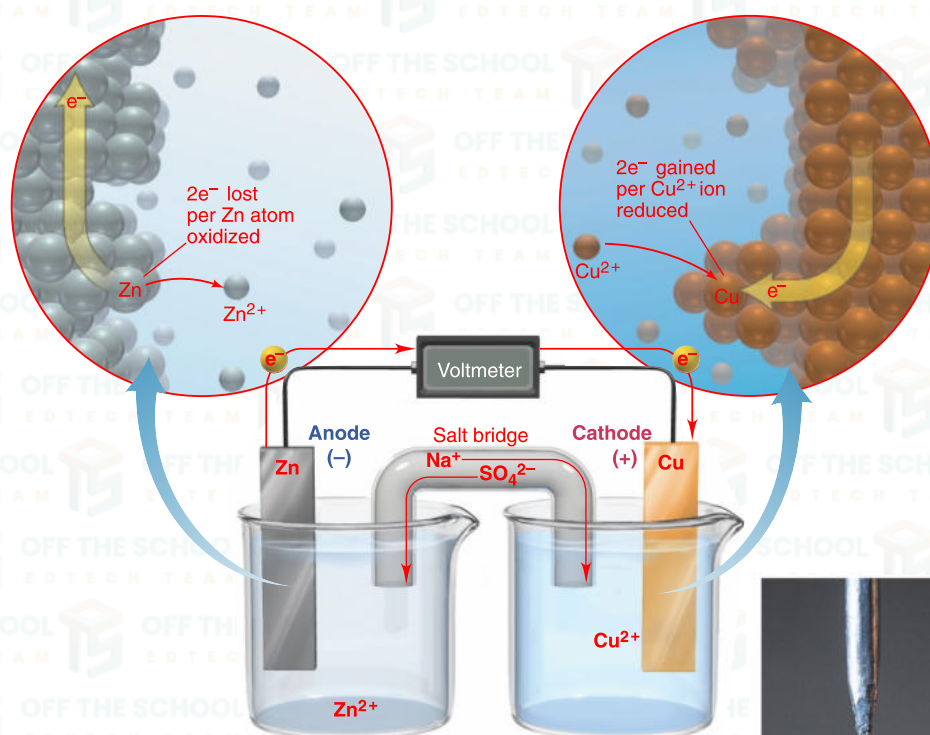
SALT BRIDGE:

A U-shaped glass tube containing an inert electrolyte that is prepared in gel which allows the flow of ions but prevents the mixing of the solution.

Working:

The electric current spontaneously flows from the Zinc electrode to the copper electrode.

Oxidation occurs at the Zn electrode serves as an anode, while reduction occurs at the Cu electrode which serves as a cathode.



A

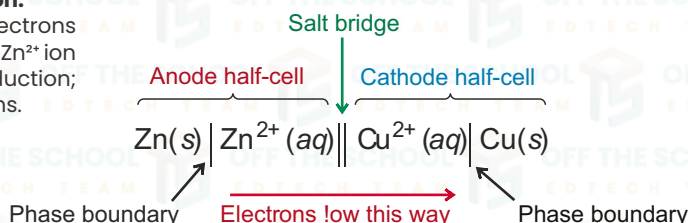
B



B, After several hours, the Zn anode weighs less and the Cu cathode weighs more.

A voltaic cell based on the zinc-copper reaction.

A, In the anode half-cell (oxidation; left), two electrons from a Zn atom move through the Zn bar as the Zn^{2+} ion enters the solution. In the cathode half-cell (reduction; right), the electrons reduce Cu^{2+} ions to Cu atoms.



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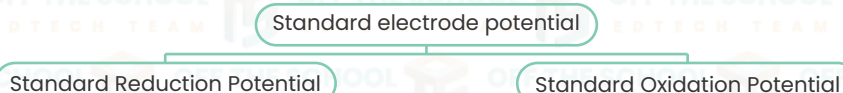
ELECTRODE POTENTIAL

CELL POTENTIAL:

The driving force with which the electrons push out from the anode into the external circuit known as electromotive force (emf) or cell potential or cell voltage.

STANDARD ELECTRODE POTENTIAL (E°):

The difference of potential created between the metal electrode & 1 molar solution of its ions at 25°C & 1 atmospheric pressure.

**STANDARD OXIDATION POTENTIAL (E°_{OXI}):**

The capability of the electrode to get oxidized in its electrolytic solution.

STANDARD REDUCTION POTENTIAL (E°_{Red}):

The capability of the electrode to gain electrons & allow positive ions from the solution to get deposited on it from the electrolytic solution

RELATION BETWEEN $E^\circ_{\text{REDUCTION}}$ & $E^\circ_{\text{OXIDATION}}$:

From the value of standard reduction potential, we can obtain the value of standard oxidation potential by just reversing the sign.

$$E^\circ_{\text{reduction}} = -E^\circ_{\text{oxidation}}$$

Example:

- Standard oxidation potential of zinc is +0.76. Therefore, its standard reduction potential is -0.76

$$E^\circ_{\text{reduction}} = 0.76\text{V}$$

$$E^\circ_{\text{oxidation}} = -0.76\text{V}$$

- The standard oxidation potential of copper is -0.34. Therefore, its standard reduction potential is +0.34.

$$E^\circ_{\text{reduction}} = 0.34\text{V}$$

$$E^\circ_{\text{oxidation}} = -0.34\text{V}$$

IUPAC convention recommended that electrode potential is taken as reduction potential therefore, for galvanic cells, the cathode should be of higher reduction potential than the anode.

**REFERENCE ELECTRODE:**

The electrode whose potential is known & used to determine the standard potential of the electrode of interest.

Example:

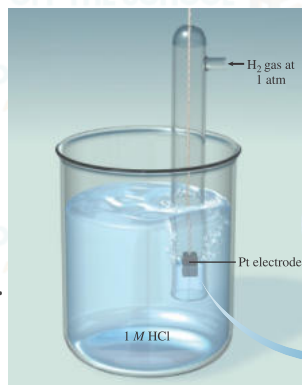
Standard Hydrogen Electrode (SHE) is the reference electrode used to determine the standard reduction potential of Cu-electrode & Zn-electrode because the electrode potential of a single electrode can't be determined without coupling it with the reference electrode.

STANDARD HYDROGEN ELECTRODE:**Structure:**

- A hydrogen electrode consists of a finely divided platinum plate dipped in a 1 molar solution of H_2SO_4 .



- At 1 atm pressure & 25°C, a pure H₂ gas is passed through the solution.
- A platinum electrode absorbs H₂ on its surface & behaves as a hydrogen electrode.
- The hydrogen electrode is designated as the standard reference electrode.
- Its standard electrode potential is 0.000 volts.



Working:

When SHE is connected with another half cell (higher reduction potential than SHE), oxidation takes place, it serves as an anode & forms H⁺ ions.



When SHE is connected with another half cell (lower reduction potential than SHE, reduction takes place, it serves as a cathode & forms H⁺ ions reduce as H₂ at the Pt electrode.



The standard potential of SHE is taken as $E^\circ_{\text{SHE}} = 0.00\text{V}$

Method for determining Standard electrode potential of Zinc:

The experiment is carried out in a galvanic cell.

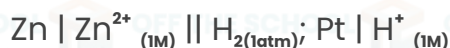
Structure or Construction of Cell:

- The galvanic cell consists of a zinc strip dipped in 1 molar Zn (NO₃)₂ or ZnSO₄.
- The standard hydrogen electrode is placed at another end.
- KCl-jelly Salt bridge is present in the middle of a cell which prevents the mixing of solutions.

EXPERIMENT & OBSERVATIONS:

- When the galvanic cell is operational, the electromotive force is found to be 0.76V.
- In the external circuit, it is observed that the flow of current takes place from the zinc to the hydrogen electrode.
- Thus, zinc is anode & negative with respect to the hydrogen electrode.

CELL NOTATION:



CELL REACTIONS:

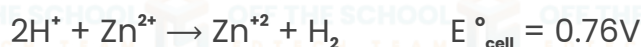
• AT ANODE Zn



• AT CATHODE SHE



CELL REACTION



CELL POTENTIAL

$$E^\circ_{\text{cell}} = E^\circ_{(\text{cathode})} - E^\circ_{(\text{anode})}$$

$$E^\circ_{\text{cell}} = E^\circ_{(\text{SHE})} - E^\circ_{(\text{Zn})}$$

$$0.76 = 0.00 - E^\circ_{(\text{Zn})}$$

$$E^\circ_{(\text{Zn})} = -0.76\text{V}$$



Method for determining Standard electrode potential of Copper:

The experiment is carried out in a galvanic cell.

Structure or Construction of Cell:

- The galvanic cell consists of a copper strip dipped in 1 molar solution of CuSO_4 .
- The standard hydrogen electrode is placed at another end. • KCl-jelly Salt bridge is present in the middle of a cell which prevents the mixing of solutions.

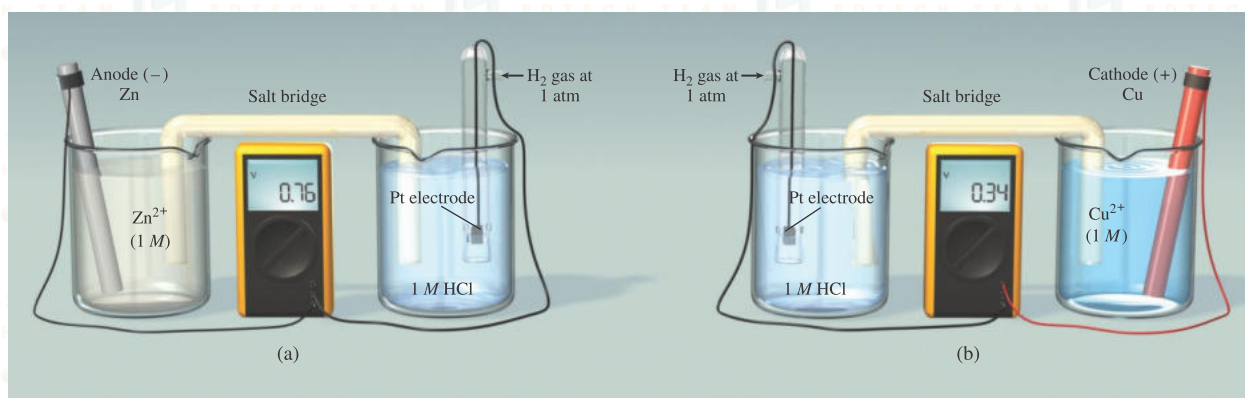
EXPERIMENT & OBSERVATIONS:

- When the galvanic cell is operational, the electromotive force is found to be 0.34V.
- In the external circuit, it is observed that the flow of current takes place from the hydrogen electrode to Cu.
- Thus, Copper reduces is cathode & positive with respect to the hydrogen electrode.

CELL NOTATION:**CELL REACTIONS:****• AT ANODE SHE****• AT CATHODE Cu****• CELL REACTION****CELL POTENTIAL**

$$\begin{aligned} E^\circ_{\text{cell}} &= E^\circ_{(\text{cathode})} - E^\circ_{(\text{anode})} \\ E^\circ_{\text{cell}} &= E^\circ_{(\text{Cu})} - E^\circ_{(\text{SHE})} \\ 0.34 &= E^\circ_{(\text{Cu})} - 0.00 \end{aligned}$$

The standard reduction potential of the Cu electrode is found to be 0.34 V.



(a) A cell consisting of a zinc electrode and a hydrogen electrode. (b) A cell consisting of a copper electrode and a hydrogen electrode. Both cells are operating under standard-state conditions. Note that in (a) the SHE is the cathode, but in (b) it is the anode.



CORROSION AND ITS PREVENTION:

The term corrosion generally refers to the deterioration of a metal by an electrochemical process. There are many examples of corrosion, including rust on iron, tarnish on silver, and the green layer that forms on copper and brass.

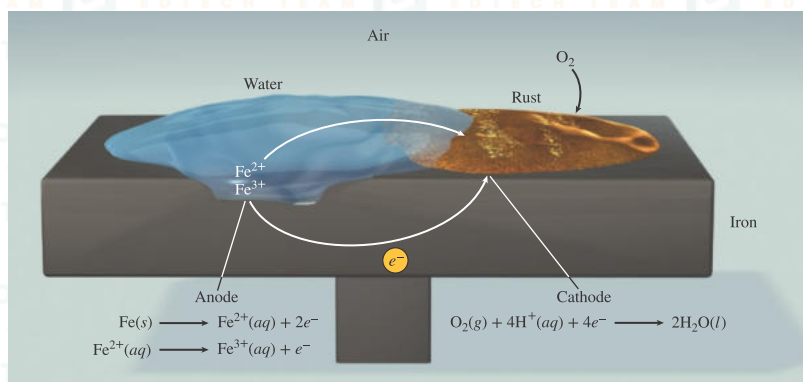
I) Rusting of iron:

Iron undergoes redox reaction in the presence of air or water to form hydrated iron (III) oxide ($\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$) called rusting of iron.

Rusting requires oxygen and water. In fact the iron is oxidised, in this reaction:



The corrosion of iron and steel has a special name: rusting.



The electrochemical process involved in rust formation. The H^+ ions are supplied by H_2CO_3 , which forms when CO_2 from air dissolves in water.

The red-brown layer is rust.

Prevention from corrosion:

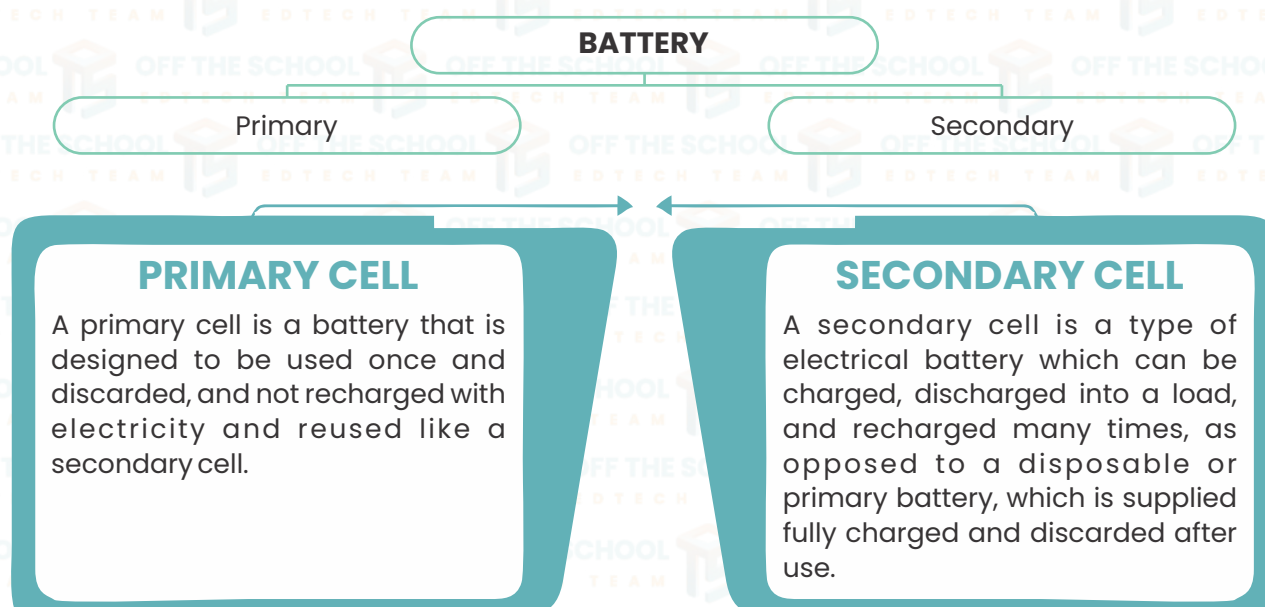
- Alloying.
- Greasing.
- Galvanizing. (Coating with zinc has a special name: galvanizing.)
- Coating with paints.
- Cathodic protection.
- Metallic coating (electroplating).
- Tin Coating.

BATTERIES:

A battery is a device that stores energy and releases it as electrical power.

Within each of these batteries are voltaic cells connected in a series that produce electrical energy.

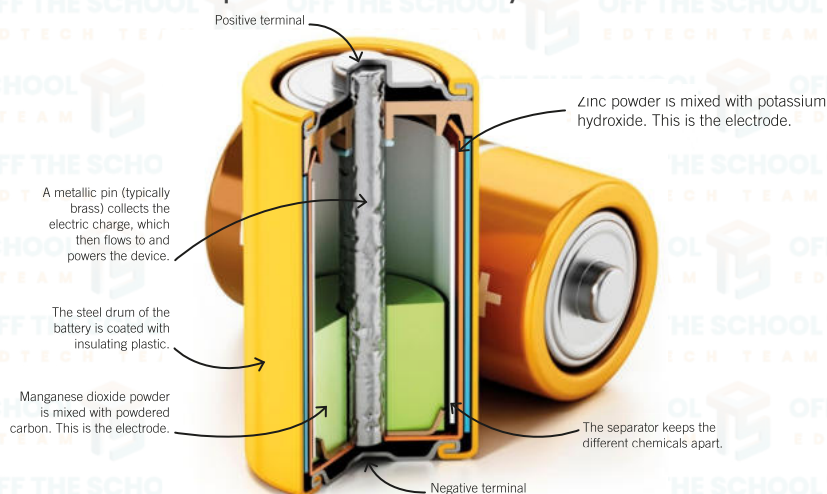
There are two types of batteries:



1) Primary (Non-rechargeable) battery. Examples include:

DRY-CELL BATTERIES:

Dry-Cell Batteries Dry-cell batteries are used in calculators, watches, flashlights, and battery operated toys. The term dry cell describes a battery that uses a paste rather than an aqueous solution. Dry cells can be acidic or alkaline.



2) Secondary (Rechargeable) battery. Example is:

LEAD STORAGE BATTERY:

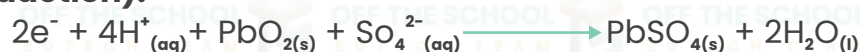
A lead storage battery is used to operate the electrical system in a car. We need a car battery to start the engine, turn on the lights, or operate the radio. If the battery runs down, the car won't start and the lights won't turn on. A car battery or a lead storage battery is a type of voltaic cell. In a typical 12-V battery, there are six voltaic cells linked together.

Each of the cells consists of a lead (Pb) plate that acts as the anode and a lead (IV) oxide (PbO₂) plate that acts as the cathode. Both half-cells contain a sulfuric acid (H₂SO₄) solution. When the car battery is producing electrical energy (discharging), the following half reactions take place:

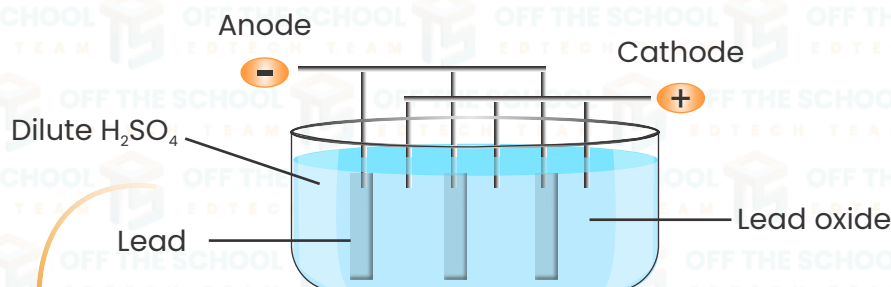
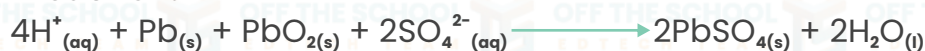
Anode (oxidation):



Cathode (reduction):



Overall cell-reaction:



Lead storage battery

